

BUFFALORA, Switzerland

WMO: 067780

Lat: 46.6481N

Lon: 10.2672E

Elev: 6463

StdP: 11.57

Time Zone: 1.00 (EUC)

Period: 11-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -10.3	(c) -5.4	(d) -16.6	(e) 2.8	(f) -9.5	(g) -11.2	(h) 3.8	(i) -3.1	(j) 15.3	(k) 24.4	(l) 13.5	(m) 24.1	(n) 2.4	(o) 130	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 24.0	(c) 73.0	(d) 53.3	(e) 70.0	(f) 52.7	(g) 67.0	(h) 51.5	(i) 55.9	(j) 66.3	(k) 54.4	(l) 65.0	(m) 53.2	(n) 63.6	(o) 7.0	(p) 290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
52.6	74.9	57.5	50.8	70.3	56.0	49.3	66.4	54.7	26.6	67.0	25.5	64.9	24.6	64.0	60.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.1	10.9	9.7	-16.7	77.5	5.3	2.4	-20.5	79.2	-23.6	80.5	-26.5	81.9	-30.4	83.6
			-17.7	58.7	5.3	1.7	-21.5	59.9	-24.6	60.8	-27.5	61.8	-31.4	63.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.1	17.0	17.8	26.0	34.0	40.5	49.2	52.1	52.2	45.1	38.1	28.4	19.8		
	DBStd	14.21	7.78	9.27	6.44	6.02	5.58	5.44	5.00	4.95	5.07	6.23	7.92	7.93		
	HDD50	5710	1024	903	744	479	297	80	36	36	159	368	649	935		
	HDD65	10912	1489	1323	1209	929	760	475	400	397	598	833	1099	1400		
	CDD50	273	0	0	0	0	3	54	101	104	11	0	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	48	0	0	0	0	0	21	15	12	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	3.7	3.5	3.6	3.9	4.1	4.2	4.1	4.0	3.8	3.5	3.6	3.6	2.9		
	PrecAvg	29.5	1.3	1.2	1.4	1.8	2.5	3.5	3.8	3.9	2.7	3.1	2.8	1.5		
(15) Precipitation	PrecMax	40.3	3.3	4.4	3.7	3.6	5.5	6.5	7.5	6.5	7.4	9.4	11.0	4.2		
	PrecMin	22.2	0.2	0.3	0.2	0.4	0.6	1.3	2.1	1.2	0.5	0.2	0.3	0.1		
	PrecStd	4.9	0.9	0.9	0.8	0.9	1.1	1.4	1.1	1.3	1.8	2.0	2.5	0.9		
	PrecStd	4.9	0.9	0.9	0.8	0.9	1.1	1.4	1.1	1.3	1.8	2.0	2.5	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.6	46.7	51.0	56.7	66.3	77.1	76.2	75.8	67.7	63.2	53.8	40.5		
		MCWB	32.8	34.3	35.9	41.6	47.5	53.4	53.9	54.0	49.6	45.4	38.7	32.7		
	2%	DB	36.2	41.1	47.2	54.0	61.3	72.0	72.3	72.6	65.1	57.5	46.8	37.0		
		MCWB	30.6	31.7	34.0	40.0	46.0	52.2	53.1	54.3	49.6	43.8	36.5	31.2		
	5%	DB	32.7	36.1	43.8	50.7	57.8	68.0	69.3	69.3	61.9	53.5	41.8	34.2		
		MCWB	29.3	29.7	32.9	38.8	44.8	51.5	51.9	53.7	48.8	42.4	35.6	30.8		
	10%	DB	30.0	31.8	40.1	47.0	53.9	63.9	66.1	66.0	58.3	49.9	38.7	31.9		
		MCWB	27.6	28.0	31.4	37.5	42.8	49.6	51.1	52.2	47.1	42.2	35.1	29.3		
	0.4%	WB	34.5	36.0	37.6	42.9	49.8	55.4	58.3	58.2	53.4	47.8	41.6	35.2		
		MCDB	38.1	42.5	46.6	51.5	60.4	67.7	67.1	69.3	61.9	56.1	48.2	37.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	32.1	32.9	35.7	41.5	47.8	53.7	56.0	56.2	51.4	45.9	39.0	33.3		
		MCDB	35.1	38.6	44.8	50.8	57.5	68.1	66.1	66.6	61.0	52.8	43.9	35.0		
	5%	WB	29.8	30.3	33.8	39.7	45.8	52.6	54.4	54.7	49.9	44.6	37.1	31.5		
		MCDB	32.1	35.2	41.1	47.7	54.1	65.8	64.6	64.8	58.5	50.0	40.3	33.4		
	10%	WB	27.6	28.3	32.2	37.7	43.9	51.0	52.9	53.4	48.6	43.3	35.3	29.3		
		MCDB	29.9	31.6	38.7	44.3	51.2	62.1	62.6	63.0	55.9	48.2	38.2	31.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	21.9	24.7	23.7	20.4	19.6	22.9	24.0	23.6	22.3	21.0	18.6	21.3		
		MCDBR	23.3	30.8	30.4	27.5	28.2	31.7	30.5	29.6	29.0	28.5	23.7	21.7		
		MCWBR	18.6	21.6	18.6	14.9	13.7	13.5	13.0	13.2	14.7	16.1	14.7	16.2		
		MCDBR	20.2	24.4	23.7	22.6	24.8	28.7	26.0	25.7	24.8	22.3	17.6	17.6		
	5% WB	MCWBR	16.7	17.6	14.7	12.9	12.9	12.7	12.3	12.4	13.4	13.8	11.8	14.1		
		MCWBR	16.7	17.6	14.7	12.9	12.9	12.7	12.3	12.4	13.4	13.8	11.8	14.1		
(40) Clear-Sky Solar Irradiance	taub		0.217	0.238	0.274	0.316	0.330	0.337	0.335	0.329	0.302	0.279	0.246	0.217		
	taud		2.371	2.320	2.294	2.313	2.380	2.399	2.403	2.439	2.503	2.544	2.509	2.408		
	Ebn at Noon		291	303	304	298	296	293	293	291	291	285	277	278		
	Edh at Noon		22	29	35	38	37	36	36	33	28	23	19	19		
(44) All-Sky Solar Radiation	RadAvg		504	806	1193	1424	1547	1776	1786	1558	1205	807	492	417		
	RadStd		56	92	149	204	164	144	140	143	121	101	78	52		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (104 neighbors)		+0.61	N/A	N/A	+1.15	+0.76	+0.65	N/A	N/A	+106	+29

Nomenclature: See separate page